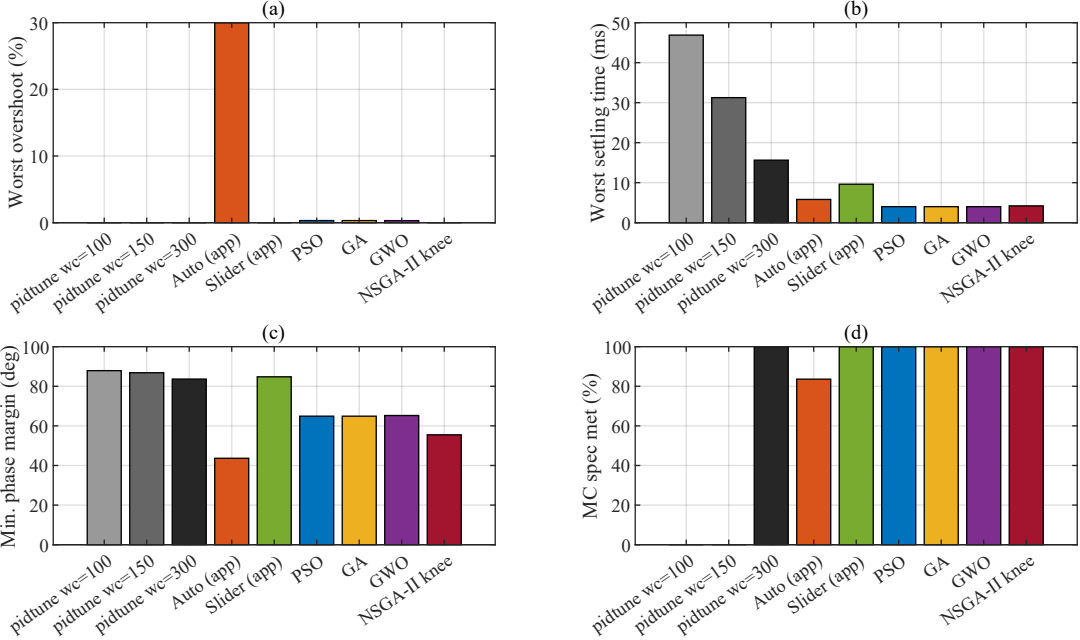

(a)
Worst overshoot (%)
30
20
10
0
pidtune wc=100
pidtune wc=150
pidtune wc=300
Auto (app)
Slider (app)
PSO
GA
GWO
NSGA-II knee
(b)
Worst settling time (ms)
50
40
30
20
10
0
pidtune wc=100
pidtune wc=150
pidtune wc=300
Auto (app)
Slider (app)
PSO
GA
GWO
NSGA-II knee
(c)
Min. phase margin (deg)
100
80
60
40
20
0
pidtune wc=100
pidtune wc=150
pidtune wc=300
Auto (app)
Slider (app)
PSO
GA
GWO
NSGA-II knee
(d)
MC spec met (%)
100
80
60
40
20
0
pidtune wc=100
pidtune wc=150
pidtune wc=300
Auto (app)
Slider (app)
PSO
GA
GWO
NSGA-II knee